

Anjali B. Yelikar

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Citizenship: India

Currently a postdoctoral scholar in the Department of Physics & Astronomy at Vanderbilt University, Nashville.

EMPLOYMENT

Vanderbilt University, TN, USA Sep 2024 | Present
GRAVITY postdoctoral scholar

Chennai Mathematical Institute, Tamil Nadu, India July 2018 | July 2019
Junior Research Fellow

EDUCATION

Rochester Institute of Technology, NY, USA Aug 2021 | Aug 2024
Ph.D. in Astrophysical Sciences and Technology
Dissertation Title: *Binary Neutron Stars: From interpreting individual events to inferring population properties inferring population properties*

Rochester Institute of Technology, NY, USA Aug 2019 | Aug 2021
MS in Astrophysical Sciences and Technology

IISER-Thiruvananthapuram, Kerala, India Aug 2013 | June 2018
BS-MS: Major in Physics, Minor in Chemistry

RESEARCH EXPERIENCE

Parameter estimation and waveform systematics in CBC signals NY, USA
Graduate researcher Aug 2019 | Aug 2024
Advisor- Dr. Richard O'Shaughnessy, Rochester Institute of Technology

Post-newtonian tests of GR Tamil Nadu, India
Junior research fellow July 2018 | July 2019
Advisor- Prof. K. G. Arun, Chennai Mathematical Institute

Fano resonance in Stimulated Brillouin Scattering (SBS) Kerala, India
Masters researcher May 2017 | May 2018
Advisor- Dr. Ravi Pant, IISER-Thiruvananthapuram

Fluorescence in organogels Kerala, India
Undergraduate researcher Jan 2017 | April 2017
Advisor- Dr. Reji Varghese, IISER-Thiruvananthapuram

Data analysis of Radio afterglows of Gamma Ray Bursts Kerala, India
Undergraduate researcher June 2016 | July 2016
Advisor- Dr. Resmi Lekshmi, IIST

Radio astronomy experiments Kerala, India
Astro club member Jan 2015 | May 2016
Project with other undergraduates, IISER-Thiruvananthapuram

RESEARCH INTERESTS

Gravitational wave astrophysics, parameter estimation, compact binary coalescences, neutron stars, extreme matter, tests of General Relativity, gamma-ray bursts

TECHNICAL SKILLS

- **Programming Languages:** Python, Mathematica, MATLAB
- **Software:** Rapid Inference via Iterative FiTting (RIFT), PESummary, Astronomical Image Processing System (AIPS)

PUBLICATIONS

Short author contributions:

- Massive Double White Dwarf Binary Mergers from the Moon: Extending the Reach of Multi-messenger Astrophysics. Manuel Pichardo Marcano, **Anjali B. Yelikar**, Karan Jani. Submitted to ApJ Letters. [arXiv:2503.04936]
- Properties of ‘Lite’ Intermediate-Mass Black Hole Candidates in LIGO-Virgo’s Third Observing Run. Krystal Ruiz-Rocha, **Anjali B. Yelikar**, Jacob Lange, William Gabella, Robert A. Weller, Richard O’Shaughnessy, Kelly Holley-Bockelmann, and Karan Jani. Submitted to ApJ Letters. [arXiv:2502.17681]
- Effects of waveform systematics on inferences of neutron star population properties and the nuclear equation of state. **Anjali B. Yelikar**, Richard O’Shaughnessy, Daniel Wysocki, Leslie Wade. Submitted to Physical Review D. [arXiv:2410.14674]
- Inference on neutron star parameters and the nuclear equation of state with RIFT, using prior EOS information. A. Vilka, **A. Yelikar**, R. O’Shaughnessy, and J. Read. [arXiv:2407.15753]
- Exploring hidden priors when interpreting gravitational wave and electromagnetic probes of the nuclear equation of state. Atul Kedia, Richard O’Shaughnessy, Leslie Wade, and **Anjali Yelikar**. [arXiv:2405.17326]
- Waveform systematics in gravitational-wave inference of signals from binary neutron star merger models incorporating higher-order modes information. **A. B. Yelikar**, R. O’Shaughnessy, J. Lange, A. Z. Jan. Physical Review D [110, 064024 (2024)]
- Low-latency parameter inference enabled by a Gaussian likelihood approximation for RIFT. **A. B. Yelikar**, V. Delfavero, and R. O’Shaughnessy. [arXiv:2301.01337]
- Improving performance for gravitational-wave parameter inference with an efficient and highly-parallelized algorithm. J. Wofford, **A. Yelikar**, Hannah Gallagher, E. Champion, D. Wysocki, V. Delfavero, J. Lange, C. Rose, V. Valsan, S. Morisaki, J. Read, C. Henshaw, and R. O’Shaughnessy. *Physical Review D* [107, 024040 (2023)] [arXiv:2210.07912]
- Surrogate light curve models for kilonovae with comprehensive wind ejecta outflows and parameter estimation for AT2017gfo. Atul Kedia, Marko Ristic, Richard O’Shaughnessy, **Anjali B. Yelikar**, Ryan T. Wollaeger, Oleg Korobkin, Eve A. Chase, Christopher L. Fryer, and Christopher J. Fontes. *Phys. Rev. Research* [5, 013168(2023)] [arXiv:2211.04363]
- General-relativistic precession in a black-hole binary. Mark Hannam, Charlie Hoy, Jonathan E. Thompson, Stephen Fairhurst, Vivien Raymond, Marta Colleoni, Derek Davis, Héctor Estellés, Carl-Johan Haster, Adrian Helmling-Cornell, Sascha Husa, David Keitel, T. J. Massinger, Alexis Menéndez-Vázquez, Kentaro Mogushi, Serguei Ossokine, Ethan Payne, Geraint Pratten, Isobel Romero-Shaw, Jam Sadiq, Patricia Schmidt, Rodrigo Tenorio, Richard Udall, John Veitch, Daniel Williams, **Anjali Balasaheb Yelikar** & Aaron Zimmerman. *Nature* [610, 652–655 (2022)].
- Compressed Parametric and Non-Parametric Approximations to the Gravitational Wave Likelihood. V. Delfavero, R. O’Shaughnessy, D. Wysocki, and **A. Yelikar**. [arXiv:2205.14154]
- Normal Approximate Likelihoods to Gravitational Wave Events. V. Delfavero, R. O’Shaughnessy, D. Wysocki, and **A. Yelikar**. [arXiv:2107.13082]
- Assessing and marginalizing over compact binary coalescence waveform systematics with RIFT. A.Z. Jan, **A.B. Yelikar**, J. Lange, and R. O’Shaughnessy. *Physical Review D* [102, 124069 (2020)]
- Testing the Kerr nature of supermassive and intermediate-mass black hole binaries using spin-induced multipole moment measurements. N. V. Krishnendu, and **Anjali B. Yelikar**. *Classical and Quantum Gravity* [37 (2020) 205019]

- Analogue of electromagnetically induced absorption in the microwave domain using stimulated Brillouin scattering. Siva Shakthi A., **Anjali B. Yelikar** and Ravi Pant. *Communications Physics* [(2020) 3:109]
- Wideband excitation of Fano resonances and induced transparency by coherent interactions between Brillouin resonances. Ravi Pant, Siva Shakthi A., **Anjali B. Yelikar**. *Scientific Reports* [2018(8): 9175]
- Everyday Radio Telescope. Pranshu Mandal, Devansh Agarwal, Pratik Kumar, **Anjali Yelikar**, Kanchan Soni, and Vineeth Krishna T. *Submitted to American Journal of Physics* [arXiv:1601.02982]

LIGO-Virgo-KAGRA collaboration paper contributions:

- GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run. *Phys. Rev. X* [13 (2023) 4, 041039]
- GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run. *Phys. Rev. D* [109, 022001 (2024)]
- Search for intermediate-mass black hole binaries in the third observing run of Advanced LIGO and Advanced Virgo. *Astronomy & Astrophysics* [659, A84 (2022)]
- Observation of Gravitational Waves from Two Neutron Star–Black Hole Coalescences. *The Astrophysical Journal Letters* [915:L5 (24pp), 2021 July 1]

PRESENTATIONS IN CONFERENCES/WORKSHOPS/SEMINARS

Talks:

- Interpreting individual events and inferring population properties from gravitational waves of coalescing binary neutron stars. **A. B. Yelikar**. *AAS Winter Meeting 2025*, 12-16 January 2025, National Harbor, MD.
- Inferring individual and population properties from gravitational waves of coalescing binary neutron stars. **Anjali Yelikar**. *Vanderbilt University Astro Seminar*, 28 Oct 2024, Nashville, TN.
- Waveform systematics in gravitational-wave inference of signals from binary neutron star merger models incorporating higher order modes information. **A. B. Yelikar**, R. O'Shaughnessy, J. Lange, and A. Z. Jan. *APS April Meeting 2024*, 3-6 April 2024, Sacramento, California.
- Waveform systematics in parameter inference of binary neutron star gravitational wave signals. *Physics Colloquium, California State University, Fullerton*, October 27, 2023.
- Testing new methods within RIFT for parameter inference of gravitational waves from NS binaries. **A. B. Yelikar**, V. Delfavero, R. O'Shaughnessy, D. Wysocki, J. Lange. *APS April Meeting 2023*, 15-18 April 2023, Minneapolis, Minnesota.
- RIFT tutorial, **A. B. Yelikar**, University of Wisconsin-Milwaukee CGCA/LSC Special Seminar, August 6, 2021.
- Effects of higher order modes in gravitational-wave inference of signals from binary neutron star mergers. **A. B. Yelikar**, A.Z. Jan, J. Lange, R. O'Shaughnessy. *APS April Meeting 2021*, 17-20 April 2021, Virtual.

Posters:

- Effects of higher order modes in gravitational-wave inference of signals from compact binaries with neutron star(s). **Anjali B. Yelikar**, Aasim Z. Jan, Jacob Lange, Richard O'Shaughnessy. *14th Edoardo Amaldi Conference on Gravitational Waves*. 19-23 July 2021, Virtual.
- Information ellipsoid volume as a figure of merit for tests of General Relativity. **Anjali B. Yelikar**, N. V. Krishnendu, Sayantani Datta, M. Saleem, K. G. Arun, B. S. Sathyaprakash. *Indian Association for General Relativity and Gravitation (IAGRG) meeting*, 3-5 January 2019, Hyderabad, India.
- A Simple Two-element Radio Interferometer. **Anjali Yelikar**, Pratik Kumar, Vineeth Krishna, Akhileshwar Mishra, Devansh Agarwal, Pranshu Mandal. *Astronomical Society of India (ASI) conference*, 10-13 May 2016, Srinagar, India.

OTHER WORKSHOPS/SCHOOLS ATTENDED

- Workshop on Gamma-ray Bursts: Prompt to Afterglow, 4-7 July 2017 NCRA, Pune, India.
- NIUS-Physics Camp, 10-20 June 2014, HBCSE-Mumbai, India.
- Vijyoshi Science Camp, 7-9 December 2013, IISc Bangalore, India.

OUTREACH

- Volunteered at RIT Observatory open house to observe the total lunar eclipse, May 15, 2022 [link]
- Exhibitor at ImagineRIT, April 23, 2022. Exhibit title- Journeys Within Space and Time: JWST and the Future of Astronomy
- Participated in the press release for two LIGO, Virgo and KAGRA collaboration papers:
 - GWTC-3, Nov 2021 [link]
 - NSBH discovery, June 2021 [link]
- Contributed to the writing of science summary for the LVK collaboration O3-IMBH paper. May 2021 [link]
- Demonstrated working of radio telescope to school students on the occasion of National Science Day at Kendriya Vidyalaya, Pattom, Kerala, India, Feb 28, 2016.